

MCARTHUR RIVER MINE, Australia

WMO: 942390

Lat: 16.442S

Lon: 136.076E

Elev: 41

StdP: 100.83

Time Zone: 9.50 (AUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	10.5	-10.0	1.6	27.6	-7.4	2.0	26.7	7.1	22.5	6.5	23.9	1.5	260	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.9	40.1	21.5	39.1	21.5	38.1	22.0	27.4	30.4	27.0	30.5	26.6	30.3	3.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.7	28.0	26.3	21.8	27.9	25.8	21.2	27.9	87.3	30.2	85.3	30.5	83.8	30.3	34.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.3	5.7	DB	6.1	42.1	1.2	1.2	5.2	43.0	4.5	43.7	3.8	44.4	3.0	45.3	(4)
(5)			WB	3.3	28.6	1.3	2.0	2.4	30.0	1.6	31.2	0.9	32.3	0.0	33.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	30.2	30.0	29.6	27.9	24.8	21.9	21.7	22.9	26.7	29.6	31.1	31.3	(6)	
	DBStd	4.17	2.04	1.89	1.70	1.86	2.88	2.99	2.83	2.73	2.30	1.98	1.78	2.11	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	12	0	0	0	0	1	5	6	1	0	0	0	0	(9)	
	CDD10.0	6308	628	560	607	536	459	356	362	399	501	608	633	659	(10)	
	CDD18.3	3279	369	327	348	286	202	111	109	142	251	350	383	401	(11)	
	CDH23.3	42807	4805	4180	4317	3487	2314	1266	1390	1935	3260	4857	5398	5597	(12)	
	CDH26.7	23013	2537	2187	2209	1821	1065	466	537	905	1833	2909	3271	3274	(13)	
(14) Wind	WSAvg	2.6	2.5	2.4	2.3	2.3	2.5	2.5	2.5	2.6	2.8	3.0	2.9	2.7	(14)	
(15) Precipitation	PrecAvg	728	178	185	141	29	3	2	3	0	6	14	52	130	(15)	
	PrecMax	1177	436	661	417	183	17	18	57	1	90	95	147	435	(16)	
	PrecMin	419	10	5	13	0	0	0	0	0	0	0	4	22	(17)	
	PrecStd	218	117	135	101	37	5	4	11	0	17	23	39	95	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	39.6	39.6	37.2	35.3	33.1	33.3	35.3	37.8	40.2	41.1	42.0	(19)	
		MCWB	22.9	21.5	21.6	21.3	20.8	19.9	19.3	19.2	19.5	19.8	21.3	20.3	(20)	
	2%	DB	38.6	38.0	37.5	35.9	34.1	32.1	32.1	33.8	36.7	38.8	39.8	40.0	(21)	
		MCWB	23.4	23.2	22.6	21.3	20.3	19.3	18.3	18.6	19.5	20.0	21.2	22.1	(22)	
	5%	DB	37.2	36.7	36.1	35.0	33.0	31.0	31.1	32.7	35.6	37.8	38.7	38.7	(23)	
		MCWB	24.1	23.5	23.3	21.4	20.1	18.5	17.6	18.0	19.3	20.5	21.4	22.6	(24)	
	10%	DB	35.6	35.3	34.9	34.1	31.9	29.6	29.9	31.5	34.6	36.7	37.5	37.2	(25)	
		MCWB	24.7	24.1	23.4	21.3	19.6	17.6	16.8	16.8	18.9	20.7	21.9	23.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	27.6	27.4	26.2	25.3	22.1	22.4	22.7	25.3	26.2	27.9	28.0	(27)	
		MCDB	30.5	30.5	31.3	28.8	29.2	26.4	29.5	29.3	29.6	29.0	29.1	29.4	(28)	
	2%	WB	27.2	27.0	26.8	25.5	24.2	21.2	21.3	21.7	23.8	25.0	26.5	27.2	(29)	
		MCDB	30.7	30.7	30.7	28.8	28.4	27.0	27.5	27.8	28.6	30.0	29.7	29.6	(30)	
	5%	WB	26.9	26.6	26.4	24.9	23.4	20.4	20.3	20.7	22.9	24.4	25.8	26.6	(31)	
		MCDB	30.7	30.4	30.3	28.8	28.2	27.2	26.1	27.4	29.0	29.7	29.8	29.8	(32)	
	10%	WB	26.5	26.3	26.1	24.3	22.5	19.6	19.2	19.6	22.1	23.9	25.2	26.1	(33)	
		MCDB	30.5	30.1	29.8	28.7	27.8	26.9	25.9	26.9	28.5	29.6	30.0	30.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	10.1	12.6	14.1	15.2	16.4	17.3	16.6	14.9	13.2	10.9	(35)	
		MCDBR	12.3	12.5	12.7	13.7	14.5	15.4	16.8	17.7	17.3	16.0	15.3	13.8	(36)	
	5% WB	MCWBR	3.3	3.3	3.3	3.9	4.5	5.3	5.6	5.5	4.6	4.5	4.0	3.8	(37)	
		MCWBR	9.2	8.9	9.1	10.4	11.9	13.7	13.8	15.1	13.5	13.4	11.4	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.409	0.404	0.391	0.372	0.352	0.331	0.330	0.336	0.393	0.420	0.436	0.427	(40)	
		taud	2.477	2.492	2.522	2.532	2.516	2.535	2.524	2.490	2.356	2.316	2.326	2.398	(41)	
	Ebn at Noon	Ebn at Noon	936	932	923	905	889	892	903	927	906	906	905	919	(42)	
		Edh at Noon	118	115	108	101	96	91	94	103	126	136	137	128	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.00	6.04	5.91	5.80	5.25	4.88	5.21	6.02	6.79	7.09	6.96	6.42	(44)	
	RadStd	RadStd	0.63	0.78	0.62	0.47	0.26	0.29	0.17	0.17	0.30	0.45	0.35	0.61	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page